

Studying organisation theory, Newell's interest was similar to Simon's and together, they started chalking out a way to build a machine that could "think" in the way they defined the term "thinking". It wasn't about the neural framework of the human brain, nor was it about coming to the best possible solution in any situation. It was based on how the human brain thought – make the best decision given the information at hand and the limitations enforced.

Still, they needed to formalise this into a science. The duo set about creating programs and carrying out studies that sought to understand how humans figured things out, i.e. the psychology involved in problem-solving. This set about a revolution in cognitive science, which sparked an interest in a branch of artificial intelligence called "Cognitive Simulation" – getting machines to mimick the way the human brain works in solving problems.

Simon and Newell had a very systematic approach to this newfound branch, carrying out meticulous psychological studies and experiments to come up with patterns and techniques people used to solve problems. These were then digitised into computer programs to help artificial intelligence systems "think" for themselves.

Human logic and applying it to AI

While the world of AI was firmly changing its focus to emulating the inner workings of the human brain, the psychology of the mind is so vast that there were several different schools of thought – no pun intended. Newell and Simon preferred to focus on techniques of problem-solving, but John McCarthy, a cognitive and computer scientist at Stanford University in the US, decided to look at it a little differently and asked AI a question we all seem to have asked someone in our lives: "Don't you have any common sense?"

Common sense might be a subjective term, but the basis of what McCarthy was talking about was logic. The human mind's capacity for logic is unparalleled in nature. And yet, it wasn't formalised as an ability till Aristotle came along and started compartmentalising and defining aspects of the human psyche.

Logic can be something as simple as a syllogism (All men are mortals. All Greeks are men. So All Greeks are mortal.) or something a bit more complex like deductive reasoning. It's when it comes to reasoning that McCarthy's contributions are highlighted.

Simon and Newell were busy figuring out the technique with which humans solved problems, and coming up with ways to turn that into digital